

Tracery Designer: A Metaball-Based Interactive Design Tool for Gothic Ornaments

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Figure 1: Some design examples using our system.

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1 INTRODUCTION

This study proposes a method for creating dynamically deformable Gothic ornaments and an interactive design support system that utilizes this method. The project specifically focuses on tracery found in window frame decorations in Gothic architecture. Traditionally, motifs of Gothic tracery are designed with combinations of arcs. However, parametrically controlling these arcs to achieve continuous deformation is challenging due to the possibility of significant changes in their conditions. Therefore, instead of using arcs, this study attempts to approximate the shape of Gothic motifs using metaballs.

Although the need for research on procedural ornamentation has long been recognized [Whitehead 2010], prior studies exploring three-dimensional forms, such as architectural decorations and wall plaques, are comparatively less common. Studies on ornaments capable of deformation or shape-shifting are even rarer. While a depiction of deforming gilded relief was attempted in [Miller 1998], his

method was limited to generating ornate patterns in a fictional style. In the context of Gothic ornamentation, the procedural generation of tracery using the Generative Modeling Language (GML) has been explored [Havemann and Fellner 2004], yet their technique was restricted to generate static and simple designs.

2 METABALLS

This study utilizes metaballs, a method for generating organic shapes, to approximate the shape of each motif. The reason is that the unique ability of metaballs to merge together like mercury allows for smooth shape-shifting, a characteristic that seems beneficial for interactive control. Metaballs can be rendered by identifying iso-density areas within multiple density fields. Additionally, by determining the iso-density areas stepwise each time a metaball is added, it is possible to produce a complex curve that partially preserves the appearance of individual metaballs (Fig. 2(b)). This study applies this feature, aiming to maintain the appearance of individual motifs while also enabling their fusion. Furthermore, utilizing density shapes like regular polygons allows for the approximation of various Gothic-style motifs with a single metaball.

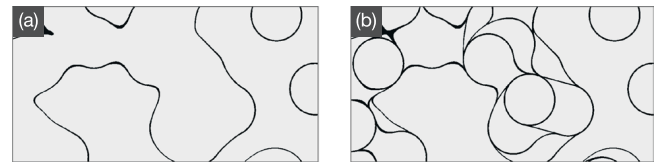


Figure 2: Regular 2D metaballs (a), and metaballs with a step-wise approach (b).

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3 MOTIF APPROXIMATION

Among Gothic ornaments, the foil is the most common motif. It can be drawn by merging partially overlapping circles in a pattern of rotational symmetry. To approximate these motifs, a single circular density distribution is set, and then the density value is attenuated at constant angular intervals. This process allows for the approximation of the motif using a single metaball, as illustrated in Fig. 3(a). By altering the angular intervals of attenuation, it becomes feasible to create various numbers of leaf decorations. Furthermore, this approach is not limited to circular shapes but can also be applied to the density distribution of regular polygons (Fig. 3(b)). Additionally, this method of density attenuation is effective in simulating the cusp, a distinctive spiky element of the Gothic style.

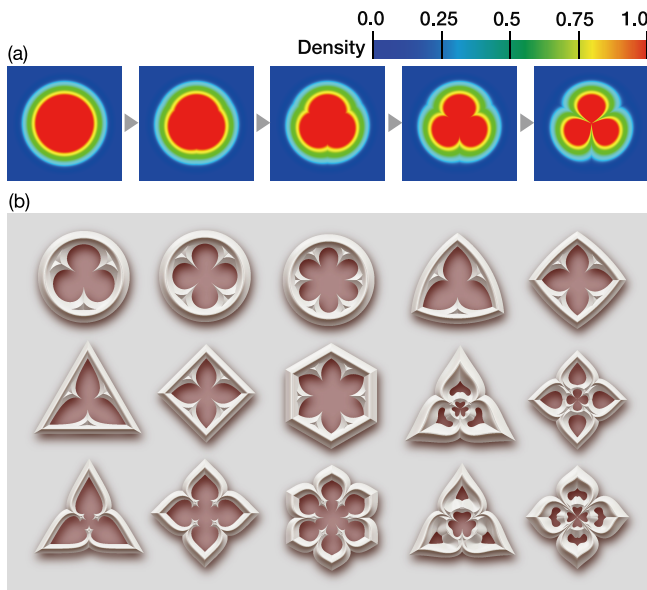


Figure 3: (a) Trefoil-looking shape by density attenuation in every 120 degrees, (b) Various motifs based on the density attenuation.

In the late Gothic style, motifs resembling a crescent moon and fish, known as mouchettes or daggers, were commonly used. To approximate these motifs, the trajectory of a metaball moving as it expands and contracts along a circular path is used as the density distribution (Fig. 4). In this case, a small radius of the moving path results in a motif with tight curves, while a radius approaching infinity leads to a straight motif.

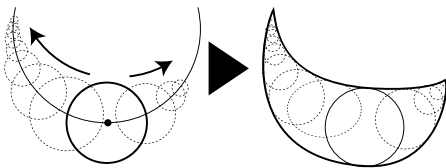


Figure 4: Concept of mouchette / dagger motifs based on the motion of a metaball.

4 IMPLEMENTATION AND ANIMATION

Using the above techniques, a prototype of an interactive design system for Gothic tracery was developed, as shown in Fig. 5. In addition to the motifs explained in the previous section, various other motifs are also implemented, such as the spire arch frequently seen in the Gothic style and the Rouleau triangle. This system, implemented in p5.js and using GLSL for graphic rendering, allows users to interactively explore various designs (Fig. 1). It features functions such as motif duplication, mirroring, and symmetrical layout capabilities. Additionally, the system can convert a design pattern into a grayscale heightmap, which is suitable for displacement mapping in external 3D software.

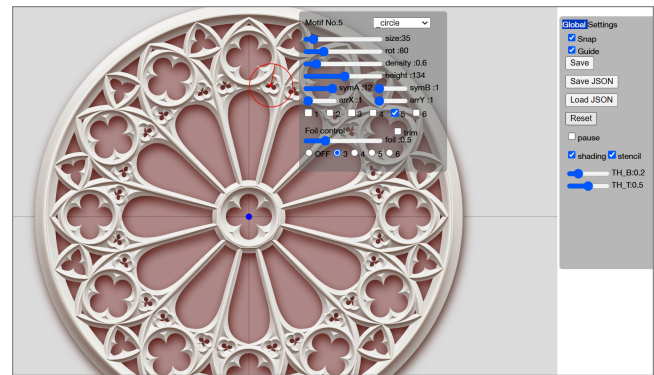


Figure 5: Screenshot of Tracery Designer.

Furthermore, to enable shape-shifting animation, a function to interpolate between two design patterns was implemented. This function makes it possible to output a sequence of images that shows the continuous transformation of the pattern. An example of such an animation, rendered in external 3D software with displacement mapping from the image sequence generated by the system, is shown in Fig. 6.



Figure 6: Example of shape-shifting animation.

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